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Citation: The Pythia citation is incorrect.

A Additional Experimental Results

A.1 Extended Distribution Analysis

This section provides additional distributional comparisons across different model architectures and sequence lengths to complement the main results.

A.2 Residual Decomposition Ablation

A.3 Hyperparameter Impact on Score Distributions

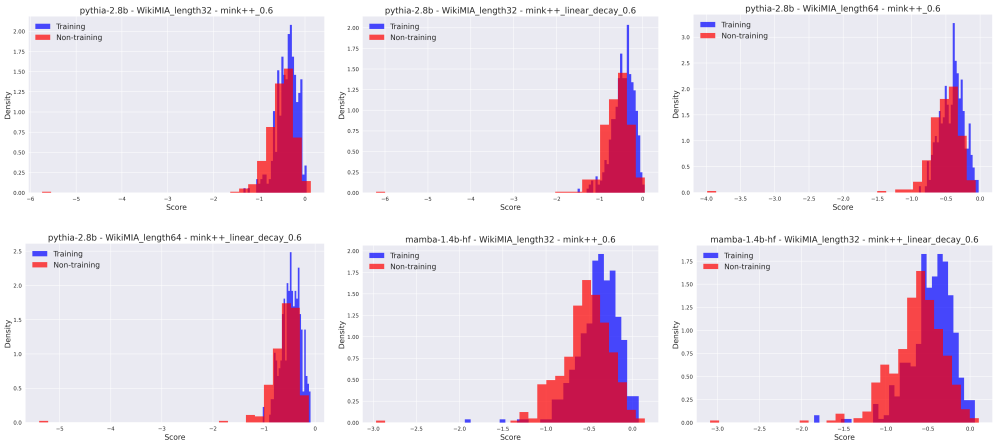

Figure 3: Comprehensive distributional analysis across models and sequence lengths. Top row: Pythia-2.8b 32-token baseline (left), proposed (center), 64-token baseline (right). Bottom row: Pythia-2.8b 64-token proposed (left), Mamba-1.4b 32-token baseline (center), proposed (right). The systematic improvements demonstrate consistent distributional enhancements across all tested configurations, with separation quality scaling with sequence length and varying by architecture.

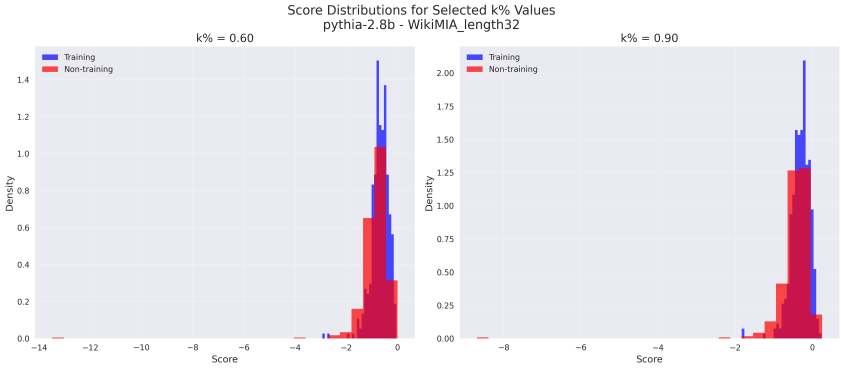

Figure 4: Distribution comparison across different Min-K ratios for Pythia-2.8b on 32-token sequences, showing how the choice of k affects the score distributions and separation quality.

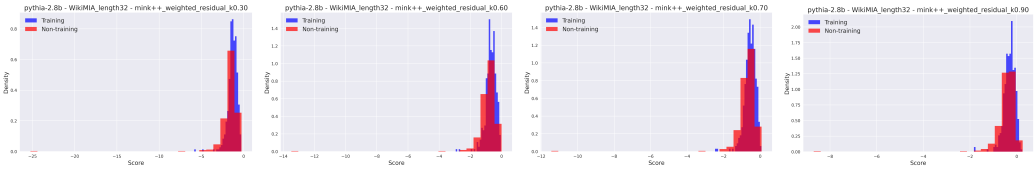

Figure 5: Score distributions across different k values (k=0.3, 0.6, 0.7, 0.9 from left to right) for Pythia-2.8b on 32-token sequences. This progression illustrates how token selection aggressiveness affects distributional characteristics: lower k values emphasize the most distinctive tokens, creating sharper separation but potentially reducing robustness, while higher k values provide broader aggregation with smoother distributions. The optimal k=0.6 balances these trade-offs effectively.